Accelerating Large-Scale Decision Tree Induction on Heterogeneous CPU-GPU Architectures

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Abstract. WIP: We investigate parallel implementations of two widely used decision tree induction algorithms, CART and C4.5, on multicore CPUs, CUDA-enabled GPUs, and heterogeneous CPU-GPU architectures.

Keywords: Decision trees · CART · C4.5 · Parallel computing

1 Introduction

Decision trees are among the most widely used models in machine learning and data mining [6]. Their predictions are easy to inspect and explain, so they suit applications in which a model must be understood or audited. They are also the base learners of random forests and gradient boosting [5,10]. CART [4] and C4.5 [18] are two of the best-known algorithms for inducing classification trees.

Growing a single tree is usually considered cheap. In practice, however, a tree is rarely trained only once. Tuning the pruning strength, depth limit or minimum leaf size needs one tree per candidate value, and k-fold cross-validation multiplies this by k. At the same time, datasets with tens of millions of samples or thousands of attributes arise in areas such as particle physics [3], bioinformatics [23] and malware detection [9]. On such data, a straightforward sequential implementation can need minutes to hours per tree, and parameter tuning pays that cost repeatedly.

Modern servers combine many cores on several sockets with high-bandwidth memory, and many are equipped with data-center GPUs. Using this hardware efficiently for tree induction is challenging, for two reasons. First, the work per node shrinks with depth. Nodes near the root hold most of the training set and provide enough work for thousands of GPU threads. The lower levels consist of many small nodes, for which kernel-launch, synchronization and transfer costs often dominate. Second, split evaluation does little computation per value read, so it is often limited by memory bandwidth.

SLIQ [14] and SPRINT [21] introduced presorting to scale to large data. Later work targeted multicore processors [2] and distributed systems [1]. Several studies proposed GPU implementations [12,16,22], but they were evaluated on GPUs of earlier generations. Most current high-performance GPU tree learners [5,10] use histogram-based split finding: continuous values are assigned to a

fixed number of bins before induction. This is cheaper, but the chosen thresholds, and therefore the trees, usually differ from those an exact CART or C4.5 search would find. To our knowledge, no public implementation grows an exact CART or C4.5 tree using the CPU and the GPU together, with each node sent to one device according to its size.

In this paper, we investigate parallel implementations of CART and C4.5 for multicore CPUs, CUDA-enabled GPUs and heterogeneous CPU–GPU systems. They search for splits exactly, without binning: every cut between two consecutive distinct attribute values is tested. All our implementations grow the same tree. We checked this node by node, against each other and against scikit-learn (CART) and Quinlan's original C4.5 program. Our speed comparisons therefore always compare runs that build the same tree.

We evaluate five implementations, from a straightforward sequential baseline to a hybrid one that uses the CPU and the GPU together. The experiments use five public datasets with up to 32 million samples and 2,381 attributes, recent CPU servers, and two generations of data-center GPUs. The optimized sequential implementation is up to 25× faster than the straightforward one. On 48 cores of a two-socket server, the multicore code is up to 40× faster again, and the GPU up to 4× faster than that. Overall, the best version is up to 1,570× faster than the straightforward sequential code. Our multicore implementation trains CART trees up to 360× faster than scikit-learn [17] and C4.5 trees up to 3,100× faster than Weka [8]. On the GPU, these factors reach 1,400× and 8,100×. For deep trees, the hybrid CPU–GPU implementation is up to 4.9× faster than growing all nodes on the GPU.

The main contributions of this paper are:

- exact parallel implementations of CART and C4.5 for multicore CPUs, GPUs and heterogeneous CPU–GPU systems, with identical trees across implementations;
- a hybrid CPU–GPU scheme that assigns the nodes of a tree to devices by size, so that the CPU and the GPU grow the tree concurrently;
- an experimental evaluation on large datasets and current hardware, covering training time and its breakdown, memory usage, scalability and energy consumption, and a comparison with established CPU and GPU implementations;
- an open-source release of the implementations.[1]

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes our implementations. Section 4 presents the experimental setup and Section 5 the results. Section 6 discusses the findings, and Section 7 concludes the paper.

[1] *[Repository URL.]*

2 Related Work

2.1 Decision Tree Induction: CART and C4.5

CART [4] builds a binary tree by recursive partitioning. At each node it chooses the split that most reduces Gini impurity, and it then simplifies the tree by cost-complexity pruning. C4.5 [18] chooses splits by gain ratio: the information gain divided by the entropy of the split itself. It splits a nominal attribute into one branch per value and a numeric attribute at a single threshold, and applies error-based pruning. For a numeric attribute, an exact search sorts the values in the node and evaluates every cut between consecutive distinct values. For entropy-based criteria, Fayyad and Irani [7] showed that an optimal cut lies on a boundary between classes in this order, so the remaining cuts can be skipped without changing the result. EC4.5 [19] obtained the same tree as C4.5 with a faster threshold search, a line continued by YaDT [20]. The cost of exact induction is therefore dominated by sorting and scanning the samples of every node, which the systems below address in different ways.

2.2 Scalable and Parallel Induction on CPUs

SLIQ [14] sorts each numeric attribute once before induction and grows the tree breadth-first. It keeps the class labels in a list that must fit in memory. SPRINT [21] removes this limit by keeping one presorted list per attribute. When a node is split, each list is partitioned between the children instead of being sorted again. SPRINT also runs on clusters, with the lists divided among the machines. Both still test every distinct value as a threshold. They scale because they avoid sorting again, not because they test fewer thresholds.

Tree induction can be parallelised in two ways. Task parallelism grows different subtrees at the same time. Data parallelism splits the work within one node, by attribute or by sample. Many systems combine both. Task parallelism suffers from load imbalance, because subtrees differ in size. Data parallelism has little work to share once nodes become small. Mixed schemes therefore use data parallelism near the root and task parallelism deeper in the tree [2]. Aldinucci et al. [2] parallelised YaDT on multicore processors with the FastFlow framework. Their NP strategy processes nodes as tasks, and their NAP strategy also splits a node's attributes across threads when a cost estimate says it pays off. They call this mix "hybrid", but on a single processor. Choosing the strategy by node size anticipates our idea of choosing the device by node size.

Yggdrasil [1] runs on a Spark cluster and splits the data by attribute. Each worker keeps all values of its attributes, sorted and, where possible, compressed. It sends back only its best split. Systems that partition by sample must exchange statistics for every candidate split, so they use binning to keep that exchange small. Yggdrasil trains deep trees on thousands of attributes without binning, but on a cluster rather than within a single machine.

2.3 GPU-Based Tree Induction

Single trees. CUDT [12] ports SPRINT's attribute lists to CUDA. The CPU controls the construction and stores the tree. The GPU evaluates the splits with prefix sums and partitions the lists, one node at a time. Its speed-up is largest at the root and shrinks with depth: nodes become smaller and more numerous, and each one needs its own host–device transfers. Nasridinov et al. [16] split the work in the same way for ID3. The tree grows one node at a time: the GPU sorts the node's samples and does the calculations, and the CPU updates the tree. They call this "hybrid CPU–GPU" and report both running time and energy on a low-end GPU. Strnad and Nerat [22] instead keep the whole construction of a CART-like binary tree on the GPU, combining node-, attribute- and split-level parallelism. Nodes are processed in batches of similar size to limit load imbalance. The schedule therefore takes node size into account, but it uses only the GPU.

Ensembles and boosting. CudaTree [11] learns random forests on the GPU. It first grows depth-first, with the whole GPU working on one large node at a time. Below a node size chosen by a fitted model, it switches to breadth-first growth, processing a whole level at once. Its authors also ran CudaTree alongside the multicore library wiseRF, but the GPU and the CPU trained different trees of the same forest rather than sharing one tree. For gradient boosting, Mitchell and Frank [15] ported the exact greedy algorithm of XGBoost [5] to the GPU. It processes all nodes of a level together, using prefix sums over the sorted attribute values. The CPU does no computation, which avoids host–device transfers but requires the training data to fit in GPU memory. ThunderGBM [24] also names the shift from a few large nodes to many small ones as a main difficulty for GPUs. Histogram-based learners such as LightGBM [10] test only bin boundaries and are not exact.

2.4 Heterogeneous CPU–GPU Induction

The works above combine the CPU and the GPU in different ways. In CUDT and in Nasridinov et al., the CPU controls the construction and the GPU computes the splits. Strnad and Nerat and CudaTree organise the work by node size, but only on the GPU. CudaTree with wiseRF is the only work above in which both devices build trees at the same time, but each device builds whole trees of a forest. Presorting, level-wise GPU growth and size-aware scheduling have all been used before. Our work differs in what each device is given, namely individual nodes of one tree, and in letting the CPU grow part of the tree rather than only control it.

In our hybrid scheme, the GPU grows the large nodes of a tree level by level. Subtrees below a size threshold go to a pool of CPU threads, which grow them while the GPU continues with the large nodes. To our knowledge, no previous work divides the nodes of a single CART or C4.5 tree between the CPU and the GPU by size, with both devices working on the tree at the same time.

3 Our Approach

3.1 Serial Simple

3.2 Serial Optimized

3.3 Multicore CPU

3.4 GPU-Based

3.5 Hybrid

4 Experiment Design

4.1 Datasets

We evaluate the implementations on five real-world datasets and one synthetic dataset, summarized in Table 1. All attributes are numeric.

SUSY and HIGGS [3] contain simulated particle collision events labelled as signal or background. Following the UCI repository, the last 500,000 samples of each serve as the test set. ECBDL'14 [23] originates from the protein structure prediction task of the ECBDL'14 Big Data Competition; its nominal attributes are encoded as integers. MNIST8M [13] consists of 8.1 million handwritten digits generated by deforming the MNIST training images, and SOREL-20M [9] is a malware detection benchmark built from Windows executables. These three datasets are used with their official test sets, which for MNIST8M is the original MNIST test set.

The training sets of ECBDL'14 and SOREL-20M (80.8 and 120.9 GB) do not fit into the 80 GB memory of our GPUs. To include them in the GPU experiments, we train on their first 15 million and 4 million samples, respectively. Both subsamples hold about $9.5 \cdot 10^9$ values, so that training uses about 73 GiB of GPU memory, and both are evaluated on the full test sets. The CPU implementations are run on the full datasets as well as on the subsamples, which allows a direct comparison with the GPU.

Chess is a synthetic two-class 3×3 checkerboard. Its two attributes are drawn uniformly from $[0, 3]$, and the class alternates between neighbouring unit squares, without noise. A tree of depth 4 with nine leaves separates the classes exactly, so training takes only a few passes over all samples. The dataset therefore measures the cost per sample, free of the overhead of growing many small nodes. We use four nested sizes from 10^6 to 10^9 samples, each a prefix of the next; the largest is the 10^8 set repeated ten times. Chess is used only to measure training time and has no test set.

4.2 Hardware

[Four nodes of the mordor3 cluster (Table 2), each used whole (exclusive jobs). GPU nodes run the CUDA backend only, CPU nodes the CPU backends and the other CPU implementations. GPU runs are bound to the GPU's NUMA node (16

Table 1. Datasets. Size: the training set as float32.

Dataset	Attributes	Classes	Training samples	Test samples	Size
SUSY [3]	18	2	4,500,000	500,000	0.3 GB
HIGGS [3]	28	2	10,500,000	500,000	1.2 GB
ECBDL'14 [23]	631	2	31,992,921	2,897,917	80.8 GB
GPU subsample	631	2	15,000,000	2,897,917	37.9 GB
MNIST8M [13]	784	10	8,100,000	10,000	25.4 GB
SOREL-20M [9]	2381	2	12,699,013	4,195,042	120.9 GB
GPU subsample	2381	2	4,000,000	4,195,042	38.1 GB
Chess 3×3	2	2	10^6–10^9		– 8 MB–8 GB

CPUs on hopper, 24 on ampere). Add measured memory bandwidth (STREAM) per node. Software: Ubuntu 24.04, Linux 7.0, g++ 13.3 (`-O3`), CUDA 12.9, NVIDIA driver 615.71.]

Table 2. Hardware. Every node has two sockets with SMT disabled; CPU, cores and L3 are per socket. Clock is base/max turbo in GHz. One GPU is used per run.

Node	CPU	Cores	Clock	L3	RAM	GPU
hopper	Xeon Gold 6444Y	16	3.6/4.0	45 MB	1 TB DDR5-4800	H100 80 GB
ampere	EPYC 7402	24	2.8/3.35	128 MB	1 TB DDR4-3200	A100 80 GB
epyc	EPYC 9255	24	3.25/4.3	128 MB	512 GB DDR5-6400	–
dell	Xeon Gold 6542Y	24	2.9/4.1	60 MB	1 TB DDR5-5200	–

4.3 Compared Implementations

[./tree backends (`-simple`, `-serial`, `-parallel`, `-cuda`, `-gpu-only`); scikit-learn and Weka J48 on the CPU; cuML and XGBoost on the GPU, with their settings. Why each is the closest equivalent (no public GPU C4.5; J48 depth limit adapter; XGBoost squared error = Gini on bins). rpart and YaDT: add or say why not.]

4.4 Protocol and Measurement

[One tree, depth $\leq$ 10, no pruning, all samples and attributes; what is timed (training only, from data in memory; GPU setup and upload included); peak RAM and VRAM; runs per case, means and spread; exclusive nodes; fairness (CPU threads of the hybrid, warm library processes, input layout). Source: docs/GPU-LIBRARIES.md, bench/README.md.]

5 Results

5.1 Training Time

[Overview of every implementation on every dataset (Tables 3 and 4): speed-ups of Serial Optimized over Serial Simple, Multicore CPU over Serial Optimized, GPU over Multicore CPU, and of this work over the existing implementations. The tables give means only (at least 10 runs per cell).]

Table 3. Training time of one CART tree (depth ≤ 10, no pruning), mean in seconds. † marks a rough estimate (no run yet); – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Multicore CPU 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs.

This work					
Dataset	Serial Simple	Serial Optimized	Multicore CPU	GPU-Based	Hybrid
SUSY	70.8	2.84	0.239	0.082	0.084
HIGGS	236	10.4	0.607	0.150	0.151
ECBDL'14	5 860	503	13.5	–	–
15M subsample	2 270	225	6.32	2.97	2.97
MNIST8M	1 880	234	5.92	3.57	3.56
SOREL-20M	12 500	878	21.8	–	–
4M subsample	2 920	234	6.43	2.87	2.87
Existing implementations					
Dataset	scikit-learn	rpart		cuML	XGBoost
SUSY	86.5	*140*†		0.133	1.43
HIGGS	215	*350*†		0.456	3.43
ECBDL'14	*4 500*†	*7 500*†		–	–
15M subsample	2 200	*3 600*†		9.95	42.2
MNIST8M	*2 500*†	*4 000*†		*7*†	*50*†
SOREL-20M	*13 000*†	*21 000*†		–	–
4M subsample	*3 500*†	*6 000*†		*10*†	*45*†

5.2 Comparison with Other CPU Implementations

[./tree's backends against scikit-learn (CART) and Weka J48 (C4.5): time, memory, test accuracy, same trees.]

5.3 Comparison with GPU Libraries

[Against cuML and XGBoost: one tree, and per tree of a 10-tree forest; accuracy of exact versus binned trees; VRAM.]

5.4 Performance Across Hardware

[All ./tree backends on the four nodes, per dataset: serial, one and two sockets, H100 and A100 hybrid and GPU-based. CPU against GPU; which hardware wins for which data shape (samples, attributes, classes).]

Table 4. Training time of one C4.5 tree (depth ≤ 10, no pruning), mean in seconds. † marks a rough estimate (no run yet); – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Multicore CPU 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs. cuML grows an entropy tree, its closest equivalent.

This work					
Dataset	Serial Simple	Serial Optimized	Multicore CPU	GPU-Based	Hybrid
SUSY	77.2	3.10	0.249	0.109	0.112
HIGGS	257	11.9	0.590	0.164	0.170
ECBDL'14	6 890	623	17.6	–	–
15M subsample	2 590	277	7.81	2.99	2.99
MNIST8M	1 970	209	5.64	3.71	3.68
SOREL-20M	12 900	1 130	28.8	–	–
4M subsample	3 090	288	7.85	3.88	4.30

Existing implementations			
Dataset	Weka J48	YaDT	cuML
SUSY	230	*46*†	0.128
HIGGS	830	*170*†	0.433
ECBDL'14	*50 000*†	*10 000*†	–
15M subsample	24 200	*4 800*†	9.89
MNIST8M	*14 000*†	*2 700*†	*7*†
SOREL-20M	*60 000*†	*12 000*†	–
4M subsample	*18 000*†	*3 800*†	*10*†

5.5 Scalability

[Thread scaling 1–48 on epyc1 and dell1 (memory bandwidth bound); growth with the number of samples (subsamples of one dataset).]

5.6 Profiling

[Where the time goes: presort, split search, partition, CPU–GPU transfer; GPU share of the hybrid; achieved bandwidth against peak.]

5.7 Energy Consumption

[Whole-server power while each implementation trains one tree on ECBDL'14 15M (the one dataset every implementation runs on, GPUs included). Method: power source and sampling rate (BMC/IPMI for the server, RAPL for the CPUs, `nvidia-smi` for the GPU), idle power measured before the runs and subtracted, energy = integral of power over the training time, 10 runs each. The CPU implementations run on a CPU node, the GPU ones on a GPU node, so the totals compare servers, not only processors. Discuss energy per tree: a fast run at high power can still use the least energy.]

Table 5. Energy of one tree on ECBDL'14 15M (depth $\leq$ 10, no pruning): mean training time t, mean power P of the whole server above idle, and energy $E = P \cdot t$. Placeholder: no measurements yet.

		CART			C4.5		
Implementation	Node	t (s)	P (W)	E (kJ)	t (s)	P (W)	E (kJ)
Serial Simple	epyc1	2 332	140	327	2 639	141	372
Serial Optimized	epyc1	225	147	33.2	278	147	40.8
Multicore CPU	epyc1	6.44	479	3.08	7.91	485	3.83
GPU-Based	hopper	2.99	316	0.95	3.03	317	0.96
Hybrid	hopper	3.01	387	1.16	3.02	394	1.19

6 Discussion

7 Conclusion

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8 TEMPORARY: Chart Style Tests

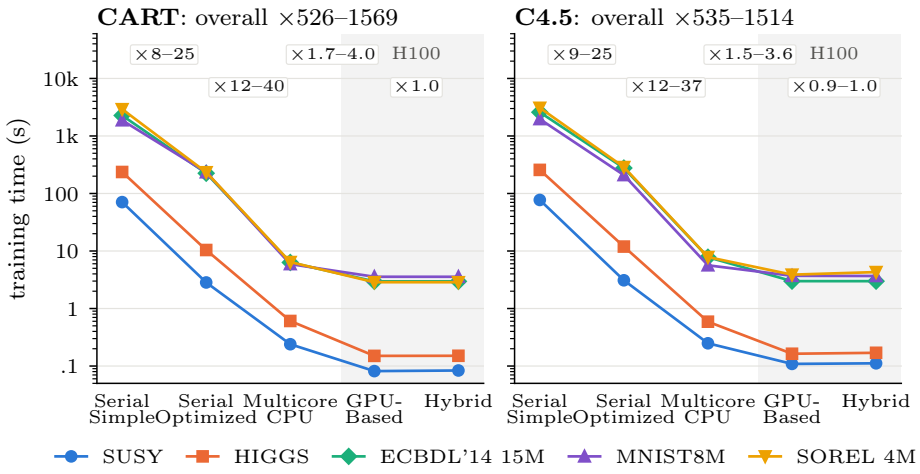


Fig. 1. Training time of one tree (depth ≤ 10, unpruned) through the steps of this work: Serial Simple, Serial Optimized and Multicore CPU on epyc1, then GPU-Based and Hybrid on the H100 (grey). Boxes: range of the step's speed-up over the five datasets.

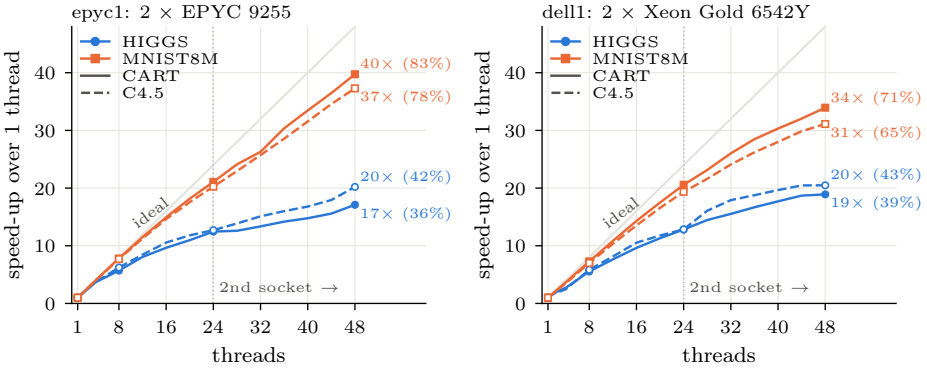


Fig. 2. Speed-up of Multicore CPU over one thread on epyc1 (left) and dell1 (right), depth ≤ 10, unpruned. Labels: speed-up and parallel efficiency at 48 threads; dotted line: the second socket starts at 25 threads.

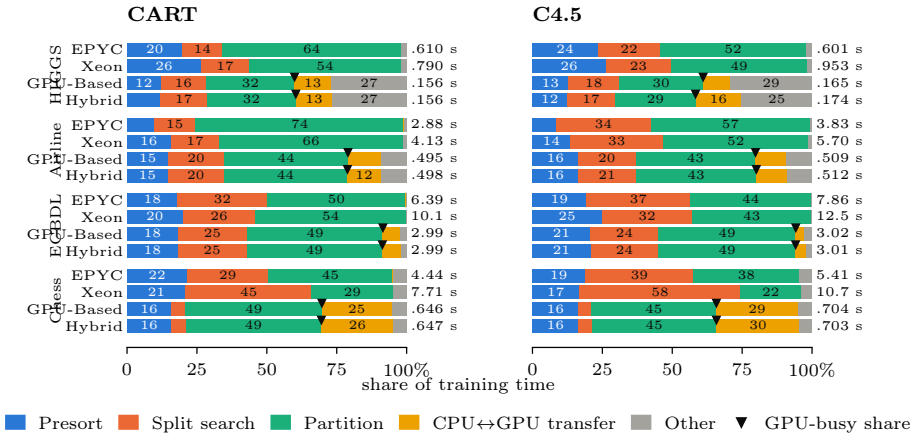


Fig. 3. Where the training time goes (percent of the training time; mean of the profiled runs, depth ≤ 10, unpruned). EPYC, Xeon: Multicore CPU with 48 threads on epyc1 and dell1; GPU-Based and Hybrid on the H100. Triangle: share of the time the GPU is busy. Chess: 3×3 board, 10^9 samples.

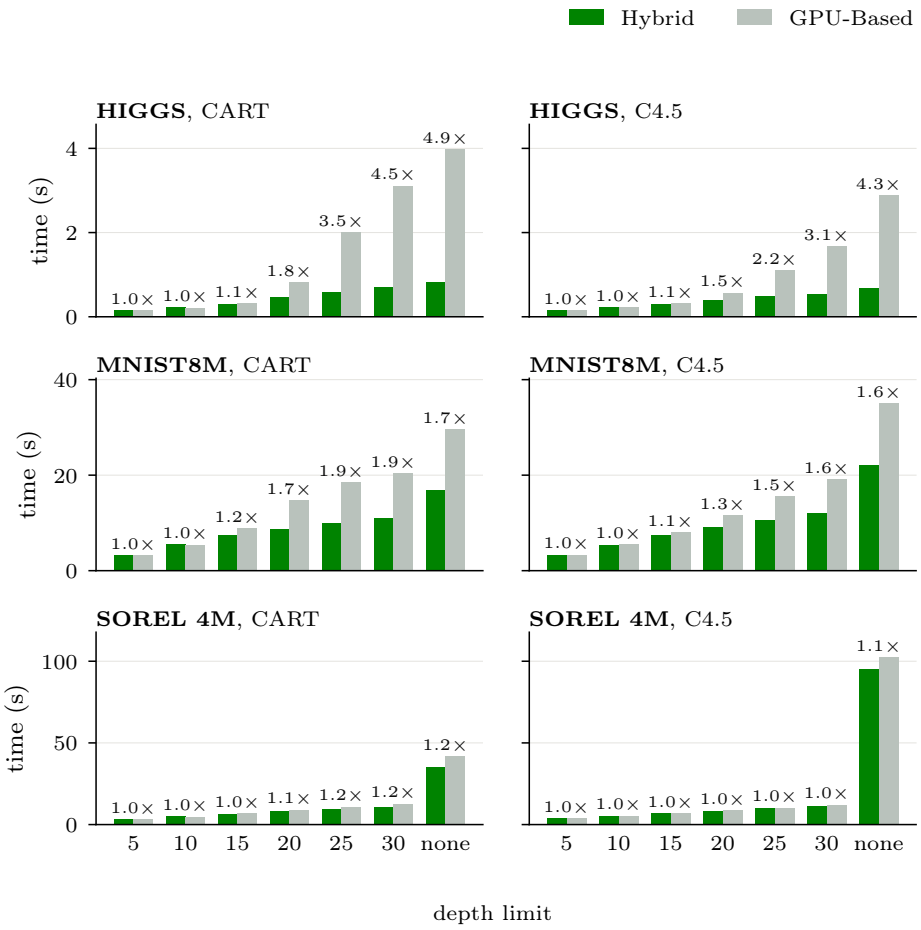


Fig. 4. Hybrid against GPU-Based by depth limit on the A100 for HIGGS, MNIST8M and the SOREL-20M 4M subsample (unpruned, mean of 3 runs, all 48 CPU threads of the node; MNIST8M at depth 5 and 10 with the 24 of the GPU's socket). Numbers above the bars: how many times slower GPU-Based is. Both modes grow the same trees.

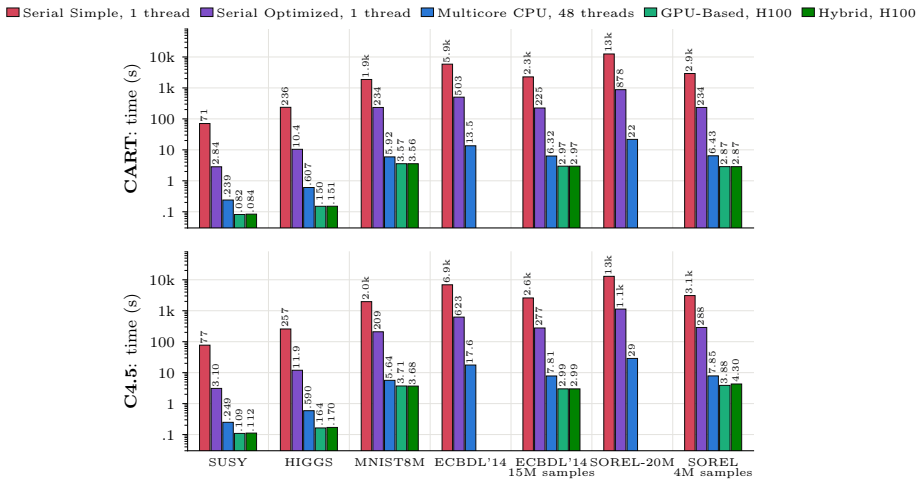


Fig. 5. Training time of one tree (depth ≤ 10, unpruned), log scale: mean over the runs. Serial Simple, Serial Optimized and Multicore CPU on epyc1, GPU-Based and Hybrid on the H100 (Hybrid with 16 CPU threads). No bar: no run (the full ECBDL'14 and SOREL-20M do not fit in the 80 GB of VRAM).

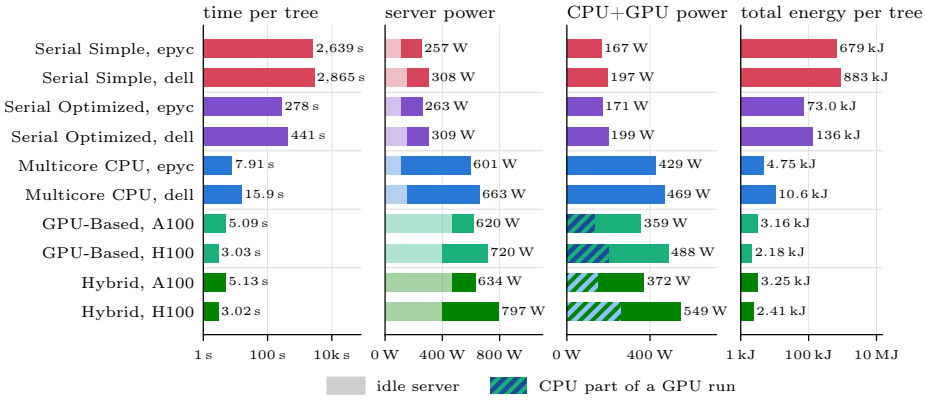


Fig. 6. Training one C4.5 tree on ECBDL'14 15M (depth ≤ 10, unpruned). Server power and energy: the whole server at the PSU, idle included. CPU+GPU power: CPU packages (RAPL) and the GPU in use (NVML). CPU implementations on epyc and dell, GPU implementations on ampere (A100) and hopper (H100).